

Ashutosh Bhardwaj

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Education

Vellore Institute of Technology | Graduating May, 2026

Vellore, Tamil Nadu

B.Tech (CSE with Data Science) | CGPA: 8.1

Technical Skills

- **Robotics & Systems:** ROS Noetic, ROS2 Humble, MoveIt, SLAM, Nav2 Stack, PID, EKF, Arduino, ESP32, CNN, YOLOv8, OpenCV
- **Programming & Tools:** Python, C++, Git, GitHub, SQL, Pandas, Matplotlib, Seaborn
- **AI & Data Science:** Power BI, TensorFlow, OCR, Object Detection, Image Preprocessing, Statistical Modeling, Regex, Text Preprocessing,

Experience

Computer Vision Intern | Oritso

May 2025 - June 2025

- Developed CNN pipeline for handwritten cheque processing, achieving **92% accuracy** on field extraction (date, amount, account number).
- Applied OCR + deep learning models (TensorFlow/Keras) to automate manual workflows, reducing cheque processing time from **10 mins to 3 mins (70% faster)**.
- Improved model robustness under noisy input by **25%**, increasing real-world deployment reliability.

Senior Autonomous Member | Team Vyadh

May 2023 - Present

- Engineered autonomous navigation using ROS, SLAM Toolbox, and Nav2, improving path efficiency by **30%** and reducing collision risk by **25%**.
- Integrated Intel RealSense D455 to generate point cloud-based 3D maps, enabling localization accuracy within **5 cm**.
- Mentored **10+ recruits**, reducing onboarding time by **40%** through structured ROS training and module-based assignments.

Projects

Autonomous Aruco-Based Navigation — [GitHub](#)

- Achieved **94% detection accuracy** for ArUco tags in varying lighting conditions using Python and OpenCV.
- Deployed model in ROS2-Gazebo to enable tag-based navigation, reducing waypoint navigation error by **35%**.

Imitation Learning Agent for Target Tracking — [GitHub](#)

- Built a behavior cloning pipeline to learn control policies from human demonstrations for autonomous navigation.
- Modeled state-action mapping using a neural network, enabling real-time target tracking in a simulated environment.

Virtual Assistant — [GitHub](#)

- Built a multi-modal AI assistant with voice and WhatsApp (Twilio) integration, supporting text, image understanding, and LLM-based conversations.
- implemented dual-memory (short-term + long-term) and a modular FastAPI backend with avatar (VTube Studio) and emotional TTS for interactive responses.

Achievements

- **International Rover Challenge 2025:** Ranked **13th** internationally among **70+ teams**.
- **International Rover Challenge 2024:** Ranked **19th** internationally.
- **NXP AIM:** Reached **Regional Finals** among **100+ competing teams**.